

Windfarm Cable Routing

Assignment 23.1

Sets

- $t \in T = \{1, 2, 3\}$: cable types

Parameters

- u_t : cost per meter of cable type t
- k_t : capacity of cable type t
- $DISTANCE_{i,j}$ distance between points i and j
- C : substation capacity
- $n + 1$: number of "points" (n turbines, 1 substation)

Decision variables

- $y_{i,j} \in \{0, 1\}$: are turbines i and j connected by a cable of any type (=1) or not (0)
- $x_{i,j}^t \in \{0, 1\}$: is cable of type t used between turbines i and j
- $f_{i,j} \geq 0$: number of turbines supported between turbines i and j

Model

Objective:

- Minimize the total cabling costs :

$$\sum_{i=1}^{n+1} \sum_{j=1}^{n+1} \sum_{t \in T} u_t \cdot DISTANCE_{ij} \cdot x_{ij}^t$$

Constraints:

- Connect cable type variables with turbine connection variables

$$\sum_{t \in T} x_{ij}^t = y_{ij} \quad \forall i = 1, 2, \dots, n+1 \quad \forall j = 1, 2, \dots, n+1$$

- Update turbines using connections into and out of a turbine

$$\sum_{j=1}^{n+1} f_{ij} - \sum_{j=1}^n f_{ji} = 1 \quad \forall i = 1, \dots, n$$

- Cable type selected must be able to support the number of connected turbines

$$\sum_{t \in T} k_t x_{ij}^t \geq f_{ij} \quad \forall i = 1, 2, \dots, n+1 \quad \forall j = 1, 2, \dots, n+1$$

- Each turbine must have one outgoing connection

$$\sum_{j=1}^{n+1} y_{ij} = 1 \quad i = 1, 2, \dots, n$$

- No flow is possible from substation

$$\sum_{j=1}^n y_{n+1,j} = 0$$

- Connection limit at substation

$$\sum_{i=1}^n y_{i,n+1} \leq C$$

The full model in Julia/JuMP is given below

```
using JuMP
using Gurobi
using HiGHS

using Plots
using GraphPlot
using LightGraphs

# number of turbines
n=34
```

```

# of cable types
T=3

# scenario 1
u=[135, 250, 480]
k=[2, 6, 9]
C=4

#= scenario 2
u=[170, 290, 610]
k=[3, 7, 13]
C=3
=#

#= scenario 3
u=[135, 290, 610]
k=[4, 6, 12]
C=3
=#

include("wind_farm_data.jl")

WFCR=Model(Gurobi.Optimizer)
#WFCR=Model(HiGHS.Optimizer)

# binary variable (cable type on edge)
@variable(WFCR, x[i=1:n+1,j=1:n+1,t=1:T], Bin)
# binary variable (edge between turbines selected or not)
@variable(WFCR, y[i=1:n+1,j=1:n+1], Bin)
# flow variable to count turbines "using" edge
@variable(WFCR, f[i=1:n+1,j=1:n+1]>=0)

# minimize cable costs
@objective(WFCR, Min, sum(u[t]*Distance[i,j]*x[i,j,t] for i=1:n+1 for j=1:n+1 for t=1:T))

# connection between x and y, if edge is selected then so is a cable type
@constraint(WFCR, [i=1:n+1,j=1:n+1], sum(x[i,j,t] for t=1:T) == y[i,j])
# turbine flow count
@constraint(WFCR, [i=1:n], sum(f[i,j]-f[j,i] for j=1:n+1) == 1)
# cable type capacity (cable type must be able to support the flow)
@constraint(WFCR, [i=1:n+1,j=1:n+1], sum(k[t]*x[i,j,t] for t=1:T) >= f[i,j])
# every turbine has one outgoing edge
@constraint(WFCR, [i=1:n], sum(y[i,j] for j=1:n+1)==1)
# no flow out of substation
@constraint(WFCR, sum(y[n+1,j] for j=1:n)==0)
# substation connection limit

```

```

@constraint(WFCR, sum(y[i, n+1] for i=1:n) <= C)

optimize!(WFCR)

if termination_status(WFCR) == MOI.OPTIMAL
    println("Cable routing costs: $(objective_value(WFCR))")
else
    println("No solution")
end

```

Assignment 23.2

New Parameters

- $ub = \lceil \frac{n}{C} \rceil$
- $lb = \lfloor \frac{n}{C} \rfloor$

New Constraints

- connections to the substation can support at most ub turbines

$$f_{i,n+1} \leq ub \quad \forall i = 1, 2, \dots, n$$

- connections to the substation must support at least lb turbines

$$f_{i,n+1} \geq lb \cdot \sum_{t \in T} x_{i,j}^t \quad \forall i = 1, 2, \dots, n$$

The full model in Julia/JuMP is given below

```

using JuMP
using Gurobi
using HiGHS

using Plots
using GraphPlot
using LightGraphs

# number of turbines
n=34
# of cable types
T=3

```

```

# scenario 1
u=[135, 250, 480]
k=[2, 6, 9]
C=4

#= scenario 2
u=[170, 290, 610]
k=[3, 7, 13]
C=3
=#

#= scenario 3
u=[135, 290, 610]
k=[4, 6, 12]
C=3
=#

include("wind_farm_data.jl")

WFCR=Model(Gurobi.Optimizer)
#WFCR=Model(HiGHS.Optimizer)

# binary variable (cable type on edge)
@variable(WFCR, x[i=1:n+1,j=1:n+1,t=1:T], Bin)
# binary variable (edge between turbines selected or not)
@variable(WFCR, y[i=1:n+1,j=1:n+1], Bin)
# flow variable to count turbines "using" edge
@variable(WFCR, f[i=1:n+1,j=1:n+1]>=0)

# minimize cable costs
@objective(WFCR, Min, sum(u[t]*Distance[i,j]*x[i,j,t] for i=1:n+1 for j=1:n+1 for t=1:T))

# connection between x and y, if edge is selected then so is a cable type
@constraint(WFCR, [i=1:n+1,j=1:n+1], sum(x[i,j,t] for t=1:T) == y[i,j])
# turbine flow count
@constraint(WFCR, [i=1:n], sum(f[i,j]-f[j,i] for j=1:n+1) == 1)
# cable type capacity (cable type must be able to support the flow)
@constraint(WFCR, [i=1:n+1,j=1:n+1], sum(k[t]*x[i,j,t] for t=1:T) >= f[i,j])
# every turbine has one outgoing edge
@constraint(WFCR, [i=1:n], sum(y[i,j] for j=1:n+1)==1)
# no flow out of substation
@constraint(WFCR, sum(y[n+1,j] for j=1:n)==0)
# substation connection limit
@constraint(WFCR, sum(y[i, n+1] for i=1:n) <= C)

```

```

# balanced
ub = ceil(n/C)
lb = floor(n/C)

@constraint(WFCR, [i=1:n], f[i,n+1] <= ub)
@constraint(WFCR, [i=1:n], f[i,n+1] >= sum(lb* x[i,n+1,t] for t=1:T) )

optimize!(WFCR)

if termination_status(WFCR) == MOI.OPTIMAL
    println("Cable routing costs: $(objective_value(WFCR))")
else
    println("No solution")
end

```

Assignment 23.3

In the solution to Scenario 3 turbine 21 is connected to turbine 13 and turbine 24 is connected to turbine 10. These connections cross each other.

New constraints

$$y_{10,24} + y_{24,10} + y_{13,21} + y_{21,13} \leq 1$$

and in general if connections (i, j) and (k, l) cross each other, the following constraint can be used

$$y_{i,j} + y_{j,i} + y_{k,l} + y_{l,k} \leq 1$$

The full model in Julia/JuMP is given below

```

using JuMP
using Gurobi
using HiGHS

using Plots
using GraphPlot
using LightGraphs

#= #number of turbines
n=34
# of cable types
T=3
=#

```

```

# scenario 1
u=[135, 250, 480]
k=[2, 6, 9]
C=4

# scenario 2
u=[170, 290, 610]
k=[3, 7, 13]
C=3
=#

# scenario 3
u=[135, 290, 610]
k=[4, 6, 12]
C=3

include("wind_farm_data.jl")

WFCR=Model(Gurobi.Optimizer)
#WFCR=Model(HiGHS.Optimizer)

# binary variable (cable type on edge)
@variable(WFCR, x[i=1:n+1,j=1:n+1,t=1:T], Bin)
# binary variable (edge between turbines selected or not)
@variable(WFCR, y[i=1:n+1,j=1:n+1], Bin)
# flow variable to count turbines "using" edge
@variable(WFCR, f[i=1:n+1,j=1:n+1]>=0)

# minimize cable costs
@objective(WFCR, Min, sum(u[t]*Distance[i,j]*x[i,j,t] for i=1:n+1 for j=1:n+1 for t=1:T))

# connection between x and y, if edge is selected then so is a cable type
@constraint(WFCR, [i=1:n+1,j=1:n+1], sum(x[i,j,t] for t=1:T) == y[i,j])
# turbine flow count
@constraint(WFCR, [i=1:n], sum(f[i,j]-f[j,i] for j=1:n+1) == 1)
# cable type capacity (cable type must be able to support the flow)
@constraint(WFCR, [i=1:n+1,j=1:n+1], sum(k[t]*x[i,j,t] for t=1:T) >= f[i,j])
# every turbine has one outgoing edge
@constraint(WFCR, [i=1:n], sum(y[i,j] for j=1:n+1)==1)
# no flow out of substation
@constraint(WFCR, sum(y[n+1,j] for j=1:n)==0)
# substation connection limit
@constraint(WFCR, sum(y[i, n+1] for i=1:n) <= C)

```

```

# balanced
ub = ceil(n/C)
lb = floor(n/C)

@constraint(WFCR, [i=1:n], f[i,n+1] <= ub)
@constraint(WFCR, [i=1:n], f[i,n+1] >= sum(lb* x[i,n+1,t] for t=1:T) )

# crossing example
@constraint(WFCR, y[10,24]+y[24,10]+y[13,21]+ y[21,13] <=1)

optimize!(WFCR)

if termination_status(WFCR) == MOI.OPTIMAL
    println("Cable routing costs: $(objective_value(WFCR))")
else
    println("No solution")
end

```